

Problem Set 3: Asset Model

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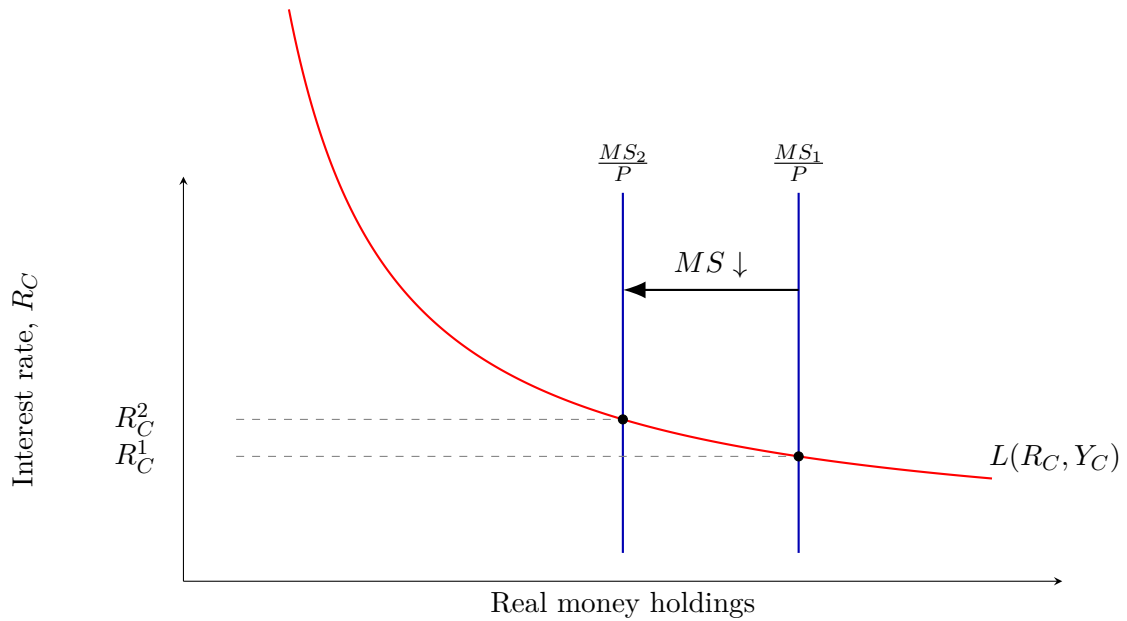
Problem 1.

(a) Policy to raise interest rates in Costa Rica.

Money market equilibrium is

$$MS = MD \iff \frac{MS}{P} = L(R_C, Y_C),$$

where R_C is the Costa Rican interest rate. In the short run P is fixed, so to increase R_C we have to decrease the real money supply $\frac{MS}{P}$. Meaning we implement a contractionary monetary policy: $MS \downarrow$.



So the policy of decreasing the supply of colóns ($MS \downarrow$) raises the interest rate ($R_C \uparrow$).

(b) Effect on the Colón/USD exchange rate.

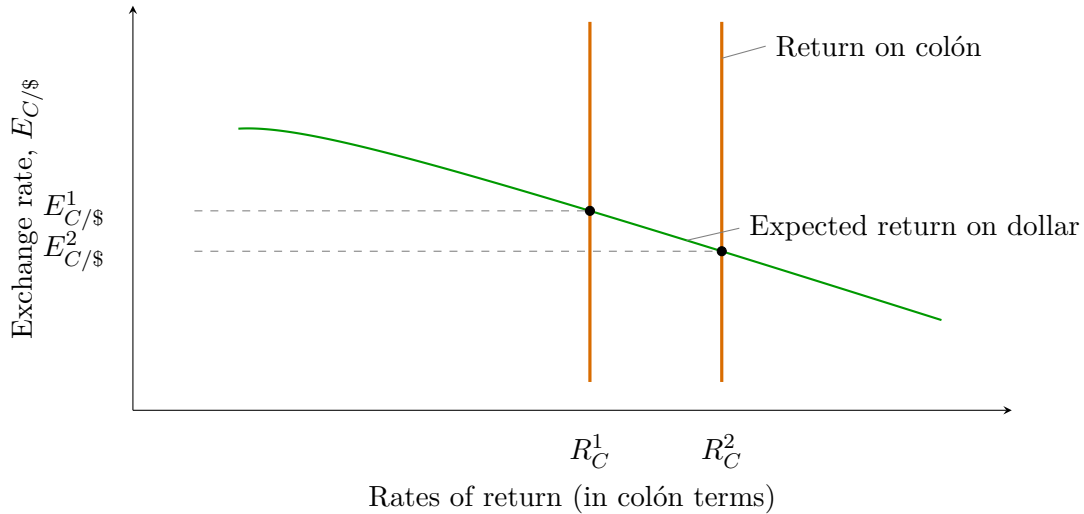
For a Costa Rican investor comparing colón deposits vs. dollar deposits, uncovered IRP is

$$\frac{1 + i_C}{1 + i_{\$}} = \frac{E_{C/\$}^{1YR}}{E_{C/\$}}.$$

Holding $i_{\$}$, $E_{C/\1YR constant, if $i_C \uparrow$ then the left-hand side rises, so $\frac{E_{C/\$}^{1YR}}{E_{C/\$}}$ must rise; with $E_{C/\1YR fixed this requires $E_{C/\$} \downarrow$.

Meaning the colón appreciates against the dollar in the short run:

$$E_{C/\$} \downarrow \iff \text{fewer colón s per \$1.}$$



Holding $i_{\$}$ and $E_{C/\1YR constant, an increase in the Costa Rican interest rate implies

$$i_C \uparrow \Rightarrow E_{C/\$} \downarrow \Rightarrow \text{the colón appreciates in the short run.}$$

(c) **Effect on probability a Costa Rican consumer buys a car produced in Costa Rica.**

We changed policy by reducing MS , which raises the Costa Rican interest rate R_C . Under the short-run assumption, prices in colón s are fixed, so the price of the Costa Rican car in colón s is unchanged in the short run.

Higher R_C increases borrowing costs and reduces the amount of financing households can obtain. Therefore the probability of purchasing the car falls:

$$MS \downarrow \Rightarrow R_C \uparrow \Rightarrow \text{Probability of buying CR car } \downarrow .$$

(d) **Effect on probability a U.S. consumer buys a Costa Rican car sold in the U.S.**

From part (b), the colón appreciates. Meaning one colón is worth more dollars, so Costa Rican goods become more expensive in dollars. The U.S. dollar price of a Costa Rican-produced car rises, lowering the probability a U.S. consumer buys it.

Holding U.S. interest rates fixed (since this policy is in Costa Rica):

$$\text{colón appreciation} \Rightarrow \text{CR goods in \$ become more expensive} \Rightarrow \text{Probability of buying CR car } \downarrow .$$

Problem 2.

Money demand is $MD = P \cdot L(R_C, Y_C)$, so an increase in income shifts $L(R_C, Y_C)$ outward:

$$Y_C \uparrow \Rightarrow L(R_C, Y_C) \text{ increases.}$$

With prices fixed and MS unchanged, money market equilibrium

$$\frac{MS}{P} = L(R_C, Y_C)$$

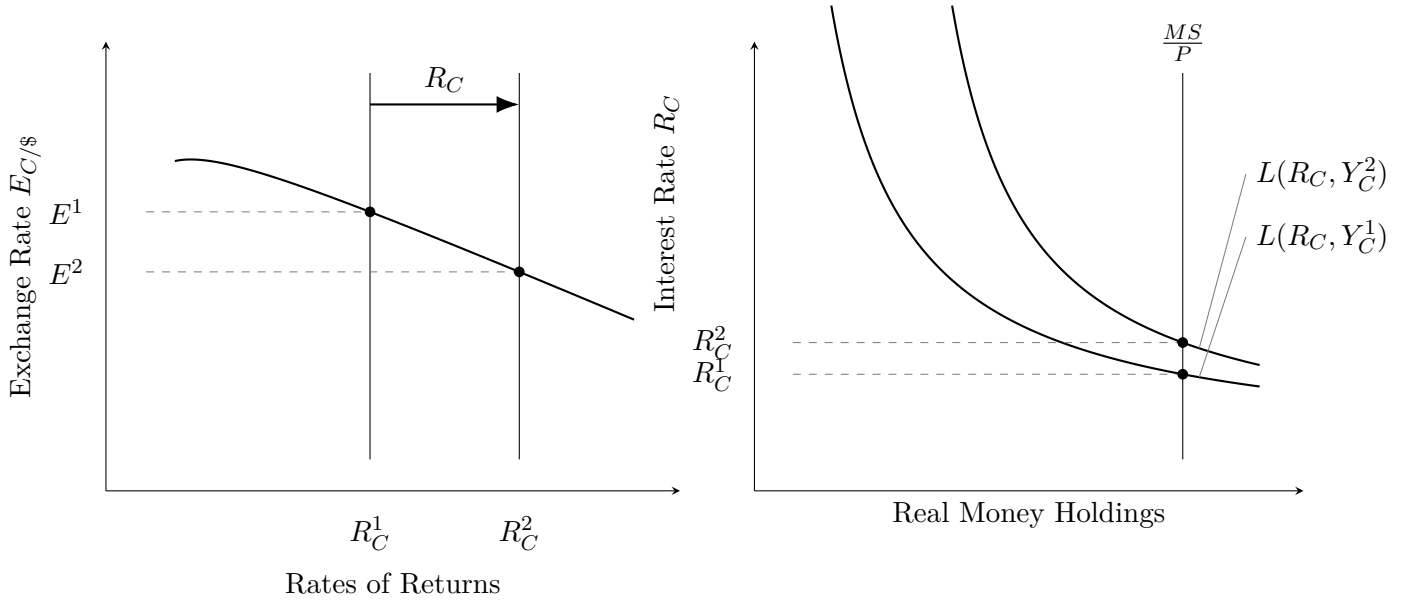
requires a higher interest rate to reduce real money demand back to the fixed supply. So:

$$Y_C \uparrow \Rightarrow L(R_C, Y_C) \uparrow \Rightarrow R_C \uparrow.$$

From IRP,

$$\frac{1 + i_C}{1 + i_{\$}} = \frac{E_{C/\$}^{1YR}}{E_{C/\$}},$$

holding $i_{\$}$, $E_{C/\1YR constant, $i_C \uparrow$ implies $E_{C/\$} \downarrow$ (colón appreciation).



So in the short-run:

$$Y_C \uparrow \Rightarrow MD \uparrow \Rightarrow R_C \uparrow \Rightarrow E_{C/\$} \downarrow \text{ (colón appreciation).}$$